

Agenda

1. Iteration problems to analyze
Tc and SC
2. Output Prediction Problems
3. Basic Debugging Problems

1. function constant()
 print("Hello World!")

How many times does the print statement execute? 1

Tc : $O(1)$

Sc : $O(1)$

2. function iterate(N)
 for i from 1 to N
 print(i)

$i = 1; i \leq N; i++$
 $i: [1 \dots N]$

How many times does the print statement execute? N

Tc : $O(N)$ Linear Tc

Sc : $O(1)$

Nested Independent

3. function analyze(N):
 for i from 1 to N
 for j from 1 to N
 print(i, j)

$i = 1; i \leq N; i++$
 N^2

for i from 1 to N
 for j from 1 to N
 print(i+1, j+1)

N^2

How many times does the print statement execute? $N^2 + N^2 = 2N^2$

TC: $O(N^2)$

SC: $O(1)$

Quadratic TC

1 val of $i \rightarrow j$ loop runs N times

N val of $i \rightarrow N \times N$

i	j	Total
1	[1...N]	N +
2	[1...N]	N +
3	[1...N]	N +
⋮	⋮	⋮ +
N	[1...N]	N
		<u>$N \times N$</u>
		$= N^2$

$$\log_2 16 = 4$$

$$2^4 = 16$$

$$2^5 \quad 2^{5.xx} \quad 2^6$$

↓ ↓ ↓

$$32 \leftarrow 35 \rightarrow 64$$

$$\log_2 35 = 5$$

$$2^5 = 32$$

How many times should I divide a number N by 2 to reach 1?

$$16 \xrightarrow[\textcircled{1}]{/2} 8 \xrightarrow[\textcircled{2}]{/2} 4 \xrightarrow[\textcircled{3}]{/2} 2 \xrightarrow[\textcircled{4}]{/2} 1 \quad \text{ans} = 4$$

$$35 \xrightarrow[\textcircled{1}]{/2} 17 \xrightarrow[\textcircled{2}]{/2} 8 \xrightarrow[\textcircled{3}]{/2} 4 \xrightarrow[\textcircled{4}]{/2} 2 \xrightarrow[\textcircled{5}]{/2} 1 \quad \text{ans} = 5$$

$$N \xrightarrow{/2} N/2 \xrightarrow{/2} N/4 \rightarrow \dots \rightarrow 1$$

$\text{ans} = \log_2 N$

4. function logScale(N)

i = 1

while i < N:

print(i)

i = i * 2

TC: $O(\log_2 N)$

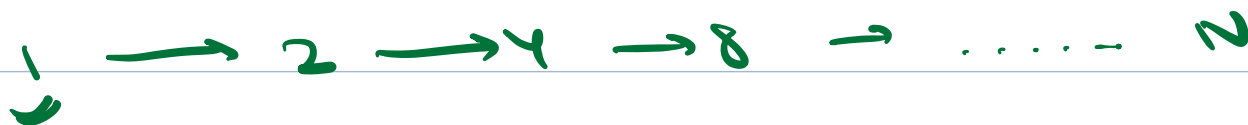
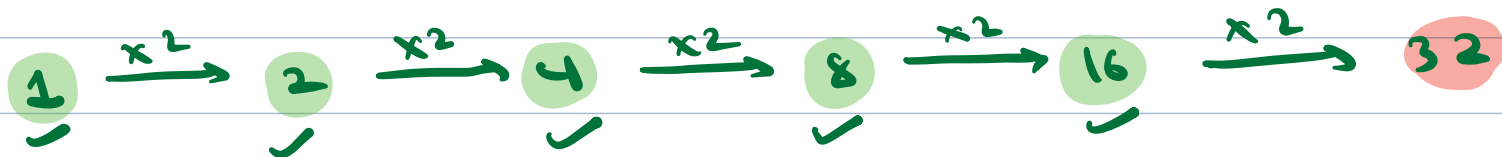
SC: $O(1)$

How many times does the print statement execute? $\log_2 N$

N = 16



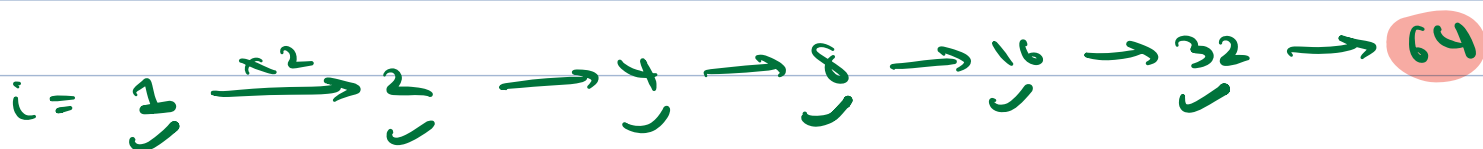
N = 32



N = 35

i < 35

i > N



$$\log_3 N < \log_2 N$$

$$\log_3 9 < \log_2 9$$

$$2 < 3 \dots$$

5. Time of algo = $(2n + 100)$ iter. Find Big O

$$\begin{array}{c} \downarrow \\ 2N^1 + 100N^0 \\ \downarrow \\ 2N^1 \rightarrow O(N) \end{array}$$

6. Time of algo = $(5n^2 + 3n + 500)$ time. Find Big O

$$\begin{array}{c} \downarrow \\ 5N^2 + 3N^1 + 500N^0 \\ \downarrow \\ 5N^2 \rightarrow O(N^2) \end{array}$$

Quiz 3

Function mysteryFunction(N)

result = 0

for i from 1 to N
 for j from 1 to N
 result = result + 1

$\sim N^2$

for i from 1 to N
 result = result + log(i)

$\sim N$

for i from 1 to 2^N
 result = result + 1

$\sim 2^N$

return result

Total iterations = $N^2 + N + 2^N$

itcs			
N	N	N^2	2^N
2	2	4	4
5	5	25	32

Higher
order term
↓
term which
grows the
fastest

TC: $O(2^N)$
SC: $O(1)$

$N < N^2 < 2^N$

Quiz 4

Function solve (N, M):

```
for i from 1 to N  
    if (N % i == 0)  
        print (i)
```

$\sum^N$ $i: [1, N]$

```
for i from 1 to M  
    if (M % i == 0)  
        print (i)
```

$\sum^M$ $i: [1, M]$

TC: $O(N+M)$

SC: $O(1)$

Quiz 5

Function func (N)

s = 0

```
for i from 1 to 100
```

$i: [1, 100]$ $\sum$ 100 iterations

s = s + i

return s

TC: $O(1)$

SC: $O(1)$

$\downarrow$
i

$s \xrightarrow{i=1, +1} 1 \xrightarrow{i=2, +2} 3 \xrightarrow{i=3, +3} 6 \xrightarrow{i=4, +4} 10 \xrightarrow{i=5, +5} \dots$

$s = 0 + 1 + 2 + 3 + 4 + \dots + 100$

$$s = \frac{\cancel{100}^{50} \times 101}{2} = 5050$$

$$S_N = \frac{N(N+1)}{2}$$

Quiz 6

Function func(N):

s = 0

i = 0

while (i < N):

s = s + i

i = i * 2

return s

i = 0 → 0 → 0 → ...
s = 0

Loop never stops

No. of iterations

are ∞

$O(\infty)$

SC: $O(1)$

Infinite Loop → TLE

Quiz 7

for i from 0 to N-1

for j from 0 to i

print (i + j)

j = 0; j ≤ i; j++

$$[a \ b] = b - a + 1$$

$$[0 \ N-1] = N-1-0+1 = N$$

Total iterations =

$$\frac{N(N+1)}{2}$$

$$= \frac{N^2 + N}{2} = \frac{N^2}{2} + \frac{N}{2}$$

TC: $O(N^2)$

SC: $O(1)$

i	[0...i]	Iterations
0	[0...0]	1 +
1	[0...1]	2 +
2	[0...2]	3
⋮	⋮	⋮
N-1	[0...N-1]	N
		<u>$\frac{N(N+1)}{2}$</u>

Quiz 8

$i=1$; $i \leq 100$; $i = i * 2$

for i from 1 to 100 step $i * 2$:

for j from 1 to N

print ($i + j$)

$i : 1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow \dots \rightarrow 100$

$\log_2 100$

$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128$

1 val of $i \rightarrow j$ loops
run N
times

	[1 N]		
	i	j	Iterations
	1	[1 N]	N
	2	[1 N]	N
	4	[1 N]	N
	8	[1 N]	N
	16		$\vdots$
	$\vdots$		

$$\frac{\log_2 100 \times N}{7N}$$

TC : $O(N)$

SC : $O(1)$

Quiz 9 which complexities grows the fastest?

$\log N$

N

$N \log N$

$N!$

$1 < \log_2 N < N < N \log_2 N < N!$

$1 < \log_2 N$

$N < N \log_2 N$

N

$$N \log_2 N$$

 $N!$

8

$$8 \log_2 8 = 24$$

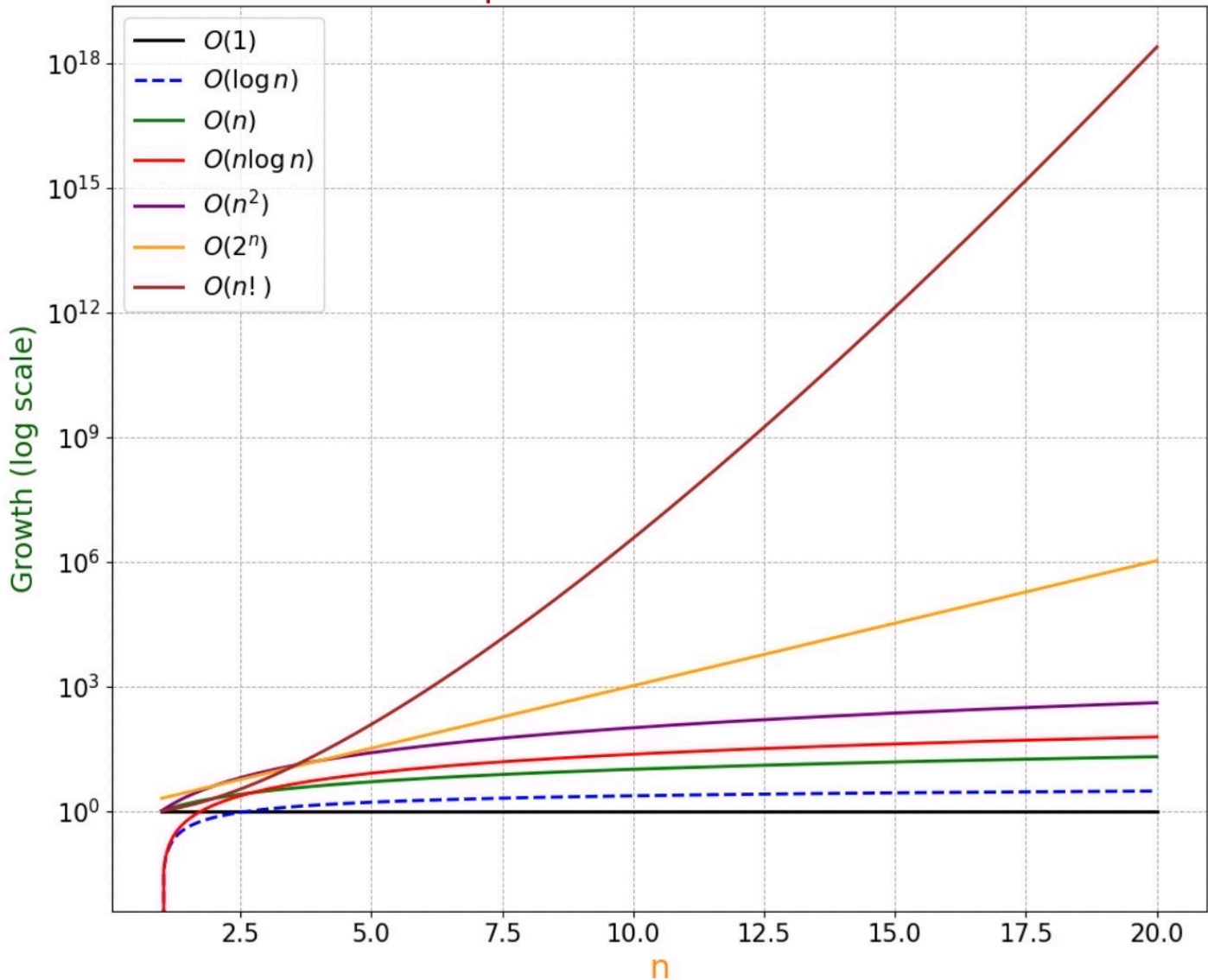
$$8! = 40320$$

16

$$16 \log_2 16 = 64$$

$$16! = 2 \times 10^{13}$$

Comparison of Growth Rates



Quiz 10: Arrange the options in the increasing order of asymptotic complexity

$$f_1(n) = n!$$

$$f_2(n) = 2^n$$

$$f_3(n) = n^n$$

$$f_4(n) = \log n$$

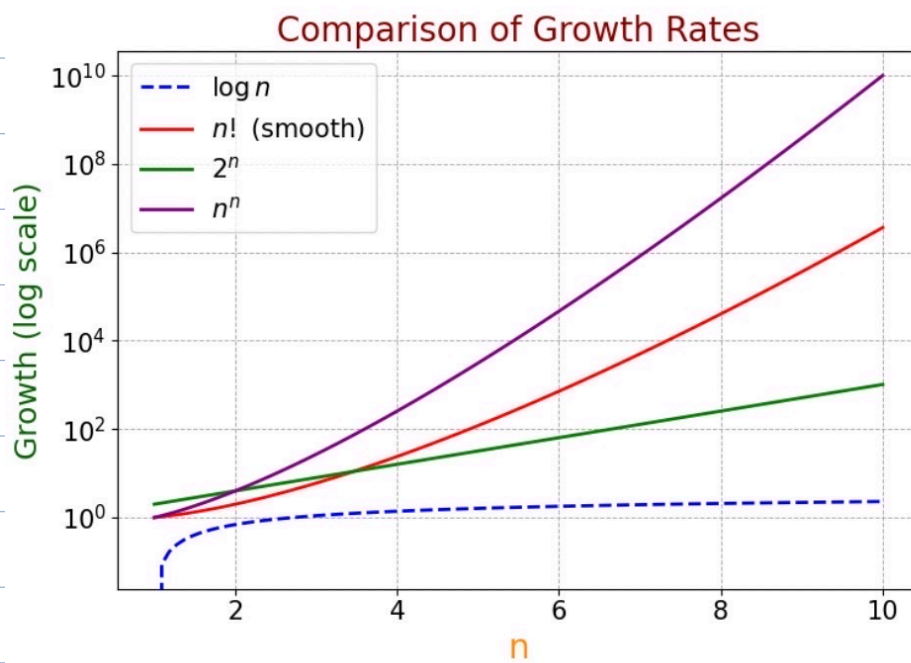
$$\log_2 n < 2^n < n! < n^n$$

$$f_4 < f_2 < f_1 < f_3$$

2
4
8

$n!$
 $4! = 24$
 $8! = 40320$

2^n
 $4^4 = 256$
 $8^8 = 1677216$



Quiz 11

$$j: 1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow \dots \rightarrow N = \log_2 N$$

Function $f_1(n)$:

result = 0

For i from 1 to n :

$j = 1$

while ($j < n$):

result = result + 1

$j = j * 2$

return result

i	j
1	$\log_2 N +$
2	$\log_2 N +$
3	$\log_2 N +$
$\vdots$	$\vdots$
N	$\log_2 N$

$$f_1 \rightarrow O(\underline{N \log_2 N})$$

Function $f_2(n)$:

result = 1

For i from 1 to n

result = result * 2

return result

$$f_2 \rightarrow O(N)$$

Function $f_3(n)$:

result = 1

For i from 1 to n

For j from 1 to n

result = result * n

return result

$$f_3 \rightarrow O(N^2)$$

Function $f_4(n)$

count = 0, $i = 1$

while $i \leq n$

count = count + 1

$i = i * 2$

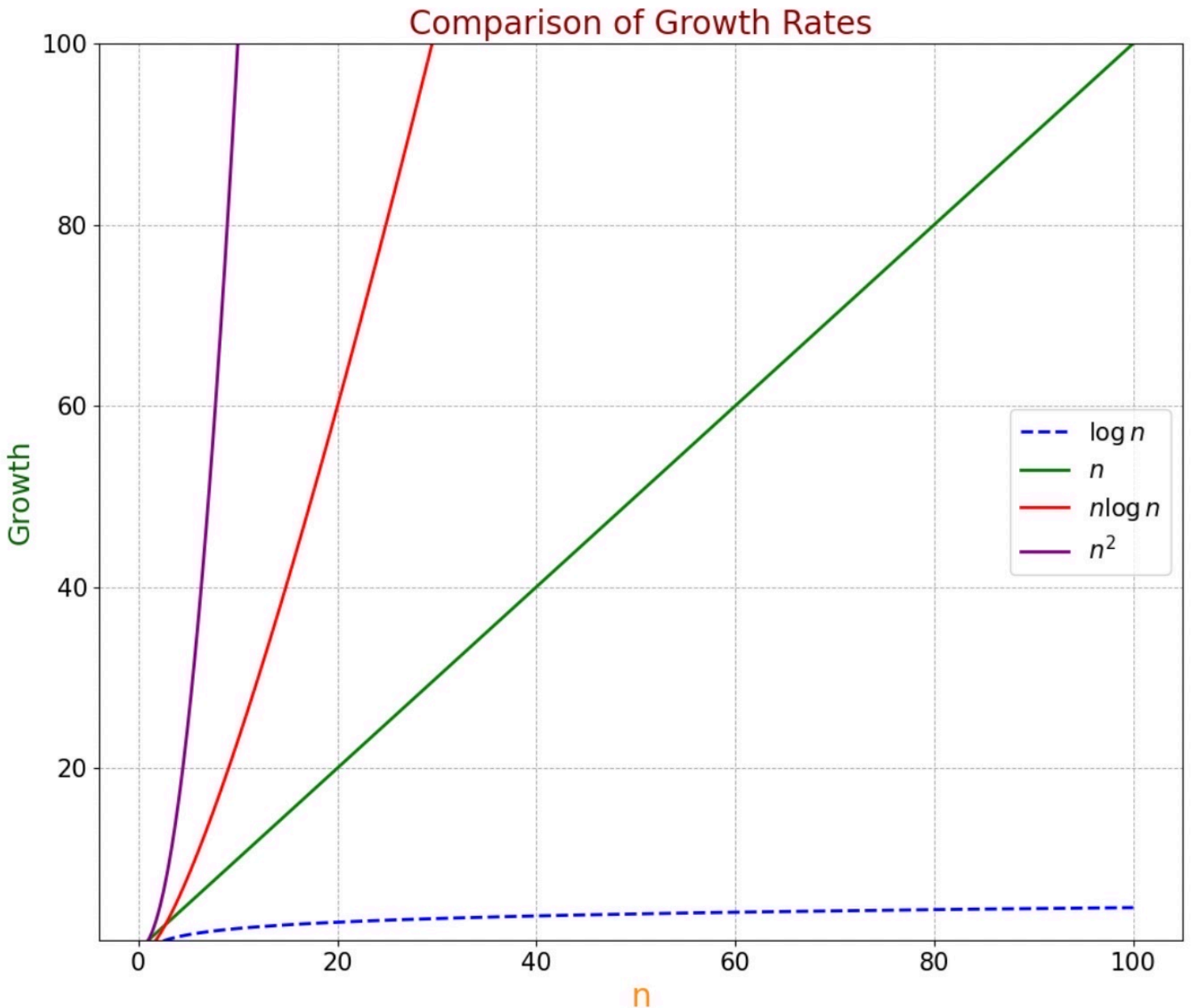
$$f_4 \rightarrow O(\log_2 N)$$

return count

Increasing order of TC ?

$$\log_2 N < N < N \log_2 N < N^2$$
$$f_4 < f_2 < f_1 < f_3$$

Break 10:50



O/P Prediction

Desmos

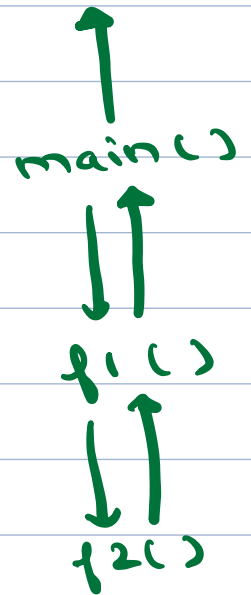
① for i from 1 to 6
if $i \% 2 == 0 \rightarrow$ is i an even no.?
print(i, "is even")

2 is even
4 is even
6 is even

② function f1()
 print("start") ✓
- f1() ✓
 print("end") ✓

function f2()
 print("Inside f2") ✓

main() <
- | f1() ✓
 ,



Start
Inside f2
End

Find and fix errors in following codes:

① `int i = 1`
`while (i < 10):`
`print(i)`

What is the issue?

∞ loop

How can we fix it? $i++$

Increment i

② `function test():`
`print(x)`
`x = 10`

What is the issue?

Undefined variable

Needs to be declared before printing

③ `i = 0 ; i ≤ N ; i++`
`for i from 0 to N:`
`print(i)`

$N = 5$

`i = 0 ; i ≤ N ; i++`
`i = 0 ; i < N ; i++`

O/P: 0 1 2 3 4 5

O/P: 0 1 2 3 4

To not print N

① $i < N$

or

② $i \leq N-1$

Quiz 12

```

int x = 10
for (int i = 1 ; i <= 5 ; i++) <
    x -= 2 // x = x - 2
    if (x % 3 == 0) <
        continue ;
    print(x)

```

O/P : 8 4 2

x = 10

i	x	x % 3 == 0
1	8	X
2	6	✓
3	4	X
4	2	X
5	0	✓
6		

Quiz 13

```

for (int i = 1 ; i <= 10 ; i += 2) <
    print(i)

```

Runs 2 no. of times

O/P : 1 3 5 7 9 11 13
15 17 ...

Quiz 14

```

for (i = 1 ; i <= 4 ; i++)
    if (i % 2 == 0 || i * 2 < 7)
        print(i)

```

i	i % 2 == 0 (i even?)	i * 2 < 7
1	X	
2	✓	✓
3	X	
4	✓	X

O/P : 2

Quiz 15

```
int i = 1
while (i < 5) <
|   print(i)
|   i++
|
>
```

Code is running
 ∞ no. of times
Throws TLE

Quiz 16 (HW: Run on IDE)

```
void find_sum() <
|   int sum
|   for (int i = 1 ; i <= 5 ; i++) <
|   |   sum += i // sum = sum + i
|   |
|   print(sum)
|
>
```

$$\text{sum} = 0 \xrightarrow[i=1]{+1} 1 \xrightarrow[i=2]{+2} 3 \xrightarrow[i=3]{+3} 6$$

Error: Sum might not have been initialized

$$0 + 1 + 2 + 3 + \dots + 5 = 15$$

Quiz 17

```
for (int i = 1 ; i <= 3 ; i++) <
|   for (int j = 3 ; j >= 1 ; j--) <
|   |   print(i + " " + j + " ")
|   |
|   print("\n")
|
>
```

$$\begin{aligned} i &= 1 & j &= 3 \\ i &= 2 & j &= 2 \\ i &= 3 & j &= 1 \\ i &= 4 & j &= 0 \end{aligned}$$

$i = 2$

$j = 3$

$j = 2$

$j = 1$

$j = 0 \times$

$$\begin{aligned} i &= 3 & j &= 3 \\ & & j &= 2 \\ & & j &= 1 \end{aligned}$$

Quiz 18

```
int[] arr = {1, 2, 3, 4}
for (i=0; i < arr.length; i++)
    print(arr[i])
```

$$i = 0; i \leq 4$$

arr of size 4
idx $\rightarrow [0, 3]$

Array Index Out of Bound Exception

Fix $i < arr.length$

Quiz 19 for ($i = 2$; $i \leq 10$; $i += 2$)
 print(i)

O/P : 2 4 6 8 10

space $\rightarrow -$

1-3-1-2-1-1-
2-3-2-2-2-1-
3-3-3-2-3-1-
C

Quiz 20 Fix: i--
for (i = 10 ; i > 0 ; i++) {
 print(i)
}

Loop is missing a decrement operator for i, leading to an ∞ loop.

Quiz 21

idx 0 1 2 3
int[] arr = {7, 5, 20, -4}
for (i = arr.length 4 ; i > 0 ; i--)
 print(arr[i])

Loop starts at arr.length(), which is out of bound.

for (i = arr.length - 1 ; i > 0 ; i--) ✓
 print(arr[i])

Drawback of normal arrays

The size has to be declared before hand.

Dynamic arrays (has automatic resizing)

It expands as you add more elements. So you don't need to determine the size ahead of time.

Java

```
ArrayList<String> a = new ArrayList<>()  
a.add("50") // 50 is inserted at end  
a.clear()  
for (int i=0 ; i<a.size(); i++) <  
|_ System.out.print(a.get(i) + " ");  
>
```

Python

```
a = []  
a.append("Orange") // Added orange at end  
a.clear()  
for i in range(len(a)):  
    print(a[i])
```

C++

```
vector<int> a  
a.push_back(60)  
a.clear()  
for (int i=0 ; i<a.size(); i++)  
    cout << a[i]
```